SHL Assessment Recommender
AI System Design

1. Objective

Build an Agentic RAG system that understands hiring intent, asks clarifications, retrieves only SHL catalog items, validates every recommendation, and returns grounded JSON responses.

2. End-to-End AI Pipeline

User → Conversation Context → Context Engineering → Intent & Entity Extraction → Clarification Decision → Embedding → Vector Search → Top-K Retrieval → Prompt Builder → LLM Reasoning → Validation → JSON Response.

3. AI Architecture

Conversation Context Layer • Context Engineering • Intent Classifier • Entity Extractor • Query Rewriter • Embedding Generator • FAISS Vector DB • Retriever • Context Ranker • Prompt Builder • LLM • Validation Agent • Output Generator.

4. Context Engineering

Conversation compression, missing-information detection, query rewriting, ambiguity resolution, conversation continuity, stateless history utilization.

5. Intent Classification

Recommendation, Clarification, Comparison, Refinement, Off-topic, Prompt Injection, Unknown.

6. Entity Extraction

Role, experience, skills, seniority, domain, soft skills, assessment preference, constraints.

7. Retrieval & Reasoning

Semantic embeddings transform user queries and catalog documents into dense vectors. FAISS retrieves Top-K relevant assessments. Context ranking prioritizes semantic similarity, skills, metadata and experience before passing grounded context to the LLM.

8. Prompt Builder

Dynamic prompt = System Instructions + Conversation History + Retrieved Catalog + Constraints + Required JSON Schema.

9. Validation Layer

Every assessment is verified against the official catalog. Invalid names or URLs are rejected. Responses remain grounded and hallucination-free.

10. AI Safety

Prompt Injection Detection • Scope Restriction • Catalog-only Knowledge • Output Validation • Refusal Policy.

11. Conversation Intelligence

Clarification strategy, comparison handling, refinement support, stateless memory, context preservation.

12. Evaluation

Schema Compliance • Recall@K • Precision • Groundedness • Hallucination Rate • Latency • Conversation Success Rate • Behavior Accuracy.

13. Failure Handling

Unknown roles, empty queries, API failures, missing catalog entries, invalid JSON, prompt injection, no matching assessment.

AI Flow Diagram

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User
↓
Conversation History
↓
Context Engineering
↓
Intent + Entity Extraction
↓
Need Clarification?
■■ Yes → Ask Question
■■ No → Embedding → FAISS → Top■K
↓
Prompt Builder
↓
LLM Reasoning
↓
Validation Agent
↓
Structured JSON
```